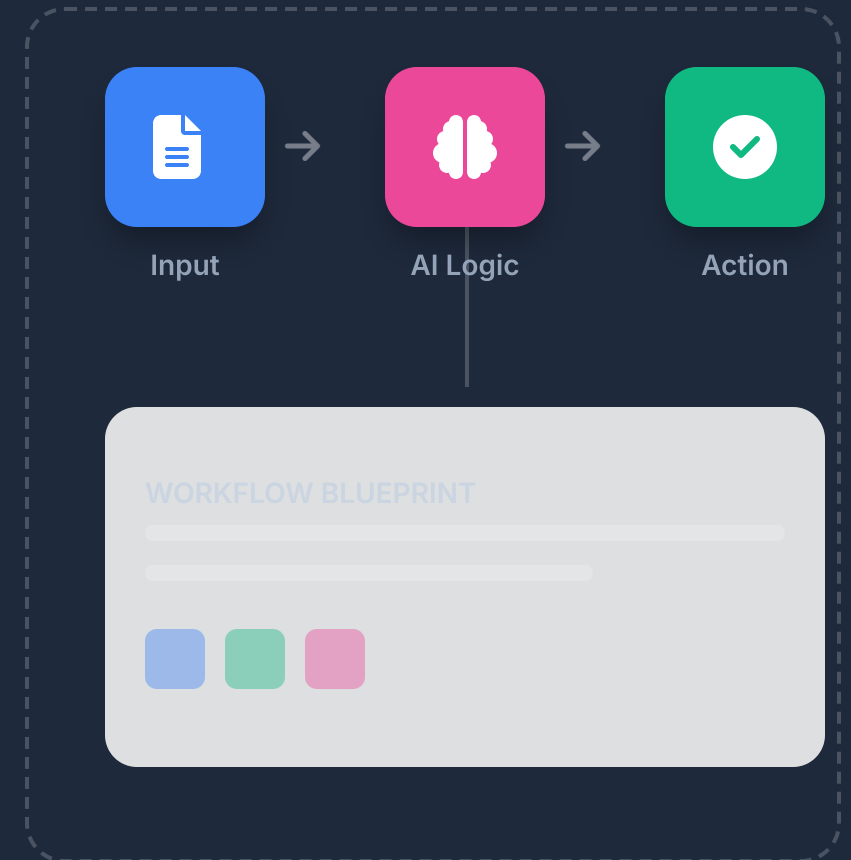


MODULE 5

Designing AI Automation Solutions

- ✓ Practical approach for non-technical teams
- ✓ Moving from ideas to working solutions
- ✓ Clean visuals and real-world patterns



SECTION OVERVIEW

What You Will Learn



Identify the right problem

Distinguish between tasks that need automation and those that don't.



Map the workflow clearly

Break down processes into triggers, actions, and conditions.



Decide where AI adds real value

Determine where interpretation, prediction, or generation is needed.



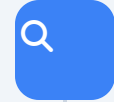
Design outputs that drive action

Create results that are measurable, useful, and actionable.



Reuse common design patterns

Apply standard patterns like "Input → AI → Output" to new problems.



DEFINITION

What Does It Mean to Design an AI Automation Solution?

It means planning how a process can be improved using a combination of four key elements:



Workflow
Structure



Automation Logic



AI Capabilities



Clear Outputs

It is not only about using a tool—it is about orchestrating these four components effectively.

Strategic Decisions Required

- 1 What problem should be solved?**
Don't automate for the sake of automation.
- 2 How should the process flow?**
Map the steps before applying technology.
- 3 Where does AI add value?**
Identify where interpretation or decision-making is needed.
- 4 What result should it deliver?**
Define the specific, measurable output.



Key Message: A strong AI automation solution starts with the **problem**, not the technology.

STEP 1

Identify the Right Problem

Not every task needs AI automation. Start with problems suitable for automation.

✔ Good Automation Candidates

- Tasks that are repetitive & high-volume
- Processes prone to human error
- Time-consuming manual work
- Dependence on manual handoffs
- Delayed by coordination steps
- Requires interpreting recurring inputs

🔍 Questions to Ask

Is this task repeated frequently?

Does it consume significant time?

Are mistakes common?

Are people doing low-value manual work?

Could AI interpret the data or text?

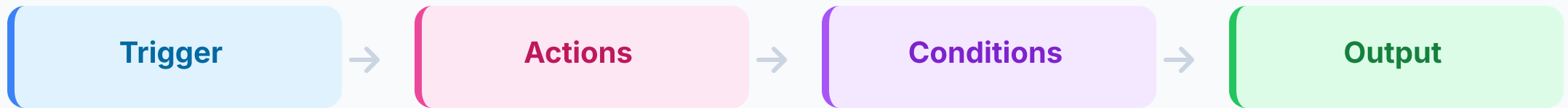
⚠ Avoid: Unclear processes, highly sensitive tasks without oversight, or very rare/low-value work.

STEP 2

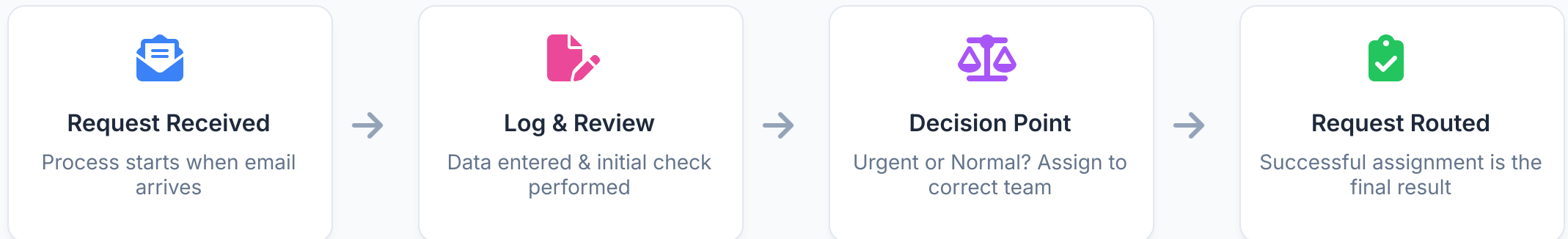
Define the Workflow Clearly

Make the process visible before automating. Map the sequence from trigger to output.

BASIC WORKFLOW MAP



PRACTICAL EXAMPLE: HANDLING REQUESTS



Key Message: You cannot design good automation without first making the workflow visible.

STEP 3

Identify Where AI Adds Value

Don't add AI everywhere. Use it where interpretation, prediction, or generation is needed.

Generation

AI CREATES:

 Summaries

 Drafts

 Reports

 Recommendations

Classification

AI CATEGORIZES:

 Request Types

 Urgency Levels

 Document Types

 Customer Intent

Prediction

AI ESTIMATES:

 Likely Outcomes

 Risks

 Priority Levels

 Future Trends

Decision Support

AI HELPS DECIDE:

 What to Review First

 Which Action to Take

 What Needs Escalation



Key Message: Use AI where **thinking or interpretation** is needed, not where simple rules are enough.

STEP 4

Define the Output Clearly

The output is the visible value of the process. It must be clear, usable, and timely.

Possible Output Types



Alerts

Instant notifications



Summaries

Condensed info



Assigned Cases

Routed to teams



Recommendations

Suggested actions



Action Triggers

System updates



Dashboard Updates

Visual reports



Design Principles



Make it clear

Outputs should be easy to read and understand without extra context.



Make it actionable

The output must clearly indicate what needs to happen next.



Connect it to the next step

Don't let the process stop. Link the output directly to the next workflow stage.



Warning: If AI summarizes data but no one receives or acts on it, the automation fails. **Output must drive action.**

Common Design Patterns for AI Automation

Across different sectors, most AI automation solutions follow these four reusable patterns.

1. Input → AI → Output



Use When: AI is mainly interpreting or generating something.
Example: Text → Summary; Request → Category.

2. Trigger → Workflow → Notification



Use When: The main need is process coordination and automation flow.
Example: Form → Review → Approval; Booking → Confirmation.

3. Trigger → AI → Decision Route



Use When: AI helps determine the workflow path.
Example: Message → AI Urgency Check → Fast-track or Standard.

4. Continuous Monitoring → Alert



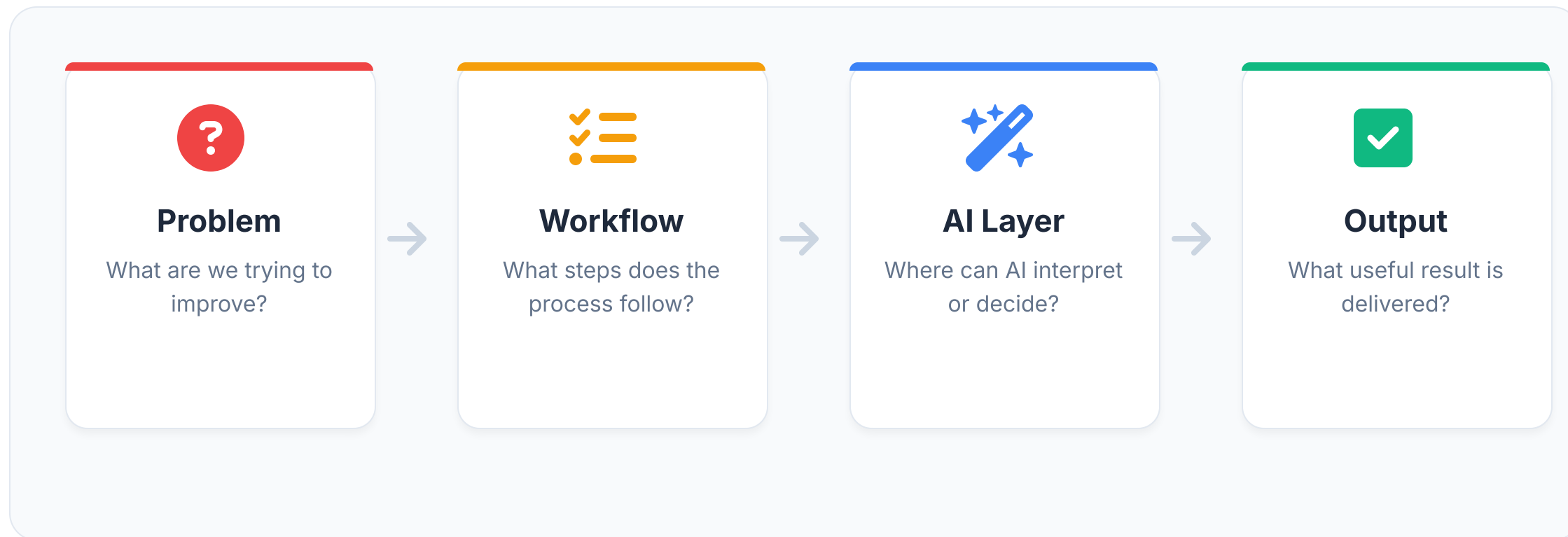
Use When: The process depends on ongoing tracking or pattern recognition.
Example: Data stream → Anomaly detection → Alert raised.

Key Message: Design patterns make automation easier to plan because they provide reusable models.

DESIGN FRAMEWORK

A Simple Design Framework to Remember

If you remember one thing from this section, use this framework to structure your automation.



Problem → Workflow → AI Layer → Output

MODULE RECAP

Section Recap and What's Next

We have covered the principles of design. Now we move to practical application.

Key Takeaways

- ✓ Start with the **problem**, not the tool.
- ✓ Map the workflow to make the **process visible**.
- ✓ Add AI only where **judgment or interpretation** is needed.
- ✓ Ensure outputs define clear **next steps or actions**.
- ✓ Reuse common **design patterns** (Input → AI → Output).



COMING UP NEXT

Practical Application

We will see these ideas come to life through **tool-based demonstrations**.

 [Healthcare Examples](#)